

Estimating Network Effects on Migrant Labor Outcomes Using a Shift-share Instrument *

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Abstract

This paper estimates the causal effects of ethnic networks on the labor outcomes of new migrants. Empirically, estimating this relationship is challenging as migrants endogenously sort into geographic regions with higher wages. Using the local concentration of migrants from the same country to proxy network effects, I construct a modified shift-share instrument to address endogeneity. I normalize the share component of the instrument by the total local immigrant population in order to reflect the predicted local coethnic concentration. I use microdata from the American Community Survey from 2005 to 2019, at the Public Use Microdata Area (PUMA) level. I construct the instrument for each origin country, year, and PUMA combination. The baseline estimates indicate inconclusive evidence for network effects on employment outcomes and wages. There is some evidence for heterogeneous effects for migrants based on origin country characteristics, occupation and industry. I also consider competition among coethnics and labor market equilibrium effects as alternative mechanisms through which local concentration of migrants affect new migrants' employment outcomes. I find that while the directional correlations are generally consistent with competition and labor market equilibrium effects, there is no conclusive evidence that any single mechanism is driving differences in new migrants' labor outcomes.

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1 Introduction

Better economic opportunities motivate international migration, despite its challenges for migrants. Migrants face frictions in new labor markets, including language and cultural barriers, unfamiliar institutions, and limited knowledge of local job market conditions. One way for migrants to overcome these issues is to gain information from fellow coethnics, migrants from the same country, who have more experience with the local labor market and institutions. Considering this, migrant enclaves naturally occur, as migrants take advantage of existing social networks to ease their transition into a new country (Bartel, 1989). However, the effects of coethnic networks on migrants' labor outcomes are ambiguous. This is because of two competing mechanisms. Coethnics can increase new migrants' wages by providing information that will reduce job search frictions, but they also compete within the same labor markets as they are most similar to each other in a local labor market. In other words, coethnics are likely to be substitutes for each other in a labor market. In addition, pure network effects could have negative effects on new migrants' wages, if the shared information constrains new migrants to low-wage labor markets. This could lead to labor market mismatches if network effects work to direct a new migrant to a job that pays lower wages than what their skills and earning potential would warrant in a frictionless market.

In this study, I ask if local ethnic networks causally affect wages and employment of new migrants. I proxy local ethnic networks by the local concentration of migrants from the same origin country. The local ethnic share acts as a measure for a migrant's potential network exposure, but it also captures labor market competition and other factors that could affect a migrant's wage.

The question of how migrants' wages are affected while working abroad is relevant to policy and to migrants themselves. With regards to policy in migrants' origin countries, this question is important as remittances have become increasingly crucial to economic growth (Goschin, 2014), even mitigating poverty (Gupta et al., 2009), increasing children's human capital gains (Bansak and Chezum, 2009) and household access to public services (Adida and Girod, 2011) in migrants' home countries. For destination countries that attract migrants, the question of how local migrant concentration affects migrant wages is relevant in order to increase labor market efficiencies and understand migrant integration into local labor markets. In addition, network effects are important

for migrants in order for them to maximize their earning potential.

Ethnic network sizes are not exogenous to a migrant’s location choice, as migrants can select and sort into local labor markets based on both wages and ethnic network sizes. To resolve this endogeneity, I use a shift-share instrument for causal identification. Shift-share instruments make use of exogenous variation in migration patterns, either via shocks in migrant inflows or historical migrant settlement patterns, allowing for causal estimation. Immigrants are defined here as individuals within the ACS who are non-citizens born outside of the U.S. Migrants’ wages and employment outcomes are the outcome variable, while the initial endogenous variable is the local proportion of immigrants from the same country. I instrument the local proportion of migrants from the same country living in an area with the shift-share. I use a representative sample of the migrant population from ACS data in this study.

I find no conclusive evidence of network effects on migrants’ wages or employment. However, I conduct heterogeneity analysis that suggests that migrants from different origin regions and origin country income status are affected differently by the local concentration of coethnics. There also seem to be heterogeneous network effects across occupations and industries, though these findings should only be interpreted as exploratory as occupations and industries could be post-treatment outcomes.

Section 2 presents the literature review on the mechanisms behind network effects affecting new migrants’ employment outcomes. Section 3 details the research design, including the use of ACS data here, and presenting and discussing the shift-share instrument I use for the estimation of network effects on migrants’ employment outcomes. Results of the baseline analysis are presented in section 4, looking specifically at network effects on new migrants’ wages and probability of employment. Section 5 presents heterogeneity analysis, checking if network effects affect migrants differently by origin country characteristics and types of jobs. Section 6 reports results from robustness checks, Section 7 discusses limitations and validity. Section 8 concludes.

2 Literature Review: Mechanisms of Network Effects on Migrants' Outcomes

Previous research has shown that local ethnic networks lead to increased probability of an immigrant being employed, increased migrant wages and welfare participation (Battisti et al., 2022; Damm, 2009; Edin et al., 2003; Bertrand et al., 2000). These studies indicate that there are network effects arising from information exchange within ethnic communication channels, connecting migrants to economic opportunities. Alternative theories are that ethnic networks provide social support (or peer pressure), human capital externalities, imposition of alternative norms, the increased difficulty for migrants to acquire skills specific to the destination country (Bertrand et al., 2000; Edin et al., 2003). These mechanisms could either positively or negatively affect new migrants' wages. Additionally, there is evidence that social networks affect employment outcomes directly through information exchange (Granovetter, 2018; Battisti et al., 2022; Bayer et al., 2008). I focus on information sharing as the main mechanism through which networks affect new migrants' labor outcomes, but other mechanisms, such as within-ethnic-group competition, could also be captured in my models of network effects on migrant employment outcomes. Since individual-level interactions and network connections are unobserved in large-scale administrative and survey datasets, the literature mainly relies on the size of local coethnic groups to proxy network exposure, which in turn proxies local network connections.

The studies on ethnic network effects affecting employment outcomes mostly use neighborhood ethnic group size controls (Battisti et al., 2022), or take advantage of policies simulating a quasi-random experiment placing immigrants into different neighborhoods (Damm, 2009; Beaman, 2012; Edin et al., 2003). However, these two approaches are limited by their internal and external validity. Using only controls produces biased estimates due to selection into neighborhoods, implying endogeneity. With regards to quasi-experimental studies, the disadvantage of using dispersal policies is that the migrants studied are largely restricted to certain countries. Though their internal validity is high due to the exogenous variation generated by political policy, their results might not generalize to the broader population of all migrants. The effect of ethnic networks plausibly differs across different demographic backgrounds, and these policies were mostly implemented in

European countries. In other words, the samples in these studies are restricted to migrants in Europe who are originally from countries that produce political migrants. These conditions might not be generalizable to the larger population of voluntary economic migrants who choose their destination locations.

To resolve these issues, I use a shift-share instrument to proxy the size of ethnic networks within a geographic area, which measures potential network effects. The shift-share instrument, first developed by Bartik (1991), uses local variation in levels of exposure to estimate the local exposure effect of a global shock to isolate the effect of an exogenous change in the independent variable on the outcome. This instrument is widely used in labor economics, especially in estimating the effects of migration on the labor market (Card, 2009; Altonji and Card, 2001). Some of the papers discussed previously use similar methods, instrumenting ethnic group size with the exogenous policy-designated location assignment for each migrant group (Beaman, 2012; Edin et al., 2003; Damm, 2009). These papers use the lagged number of migrants from the location-assignment policy as their instrument, representing an exogenously allocated share. In this paper, I use a shift-share instrument based on baseline coethnic shares instead of policy-assigned migration.

In short, I contribute to the current literature by estimating the effects of the local concentration of coethnics on migrant labor outcomes with a comprehensive sample of migrants from a large-scale dataset. My analysis relies on exogenous variation via the instrument, addressing the selection issue. It also covers different migrant populations across different local labor markets in the U.S., unlike studies exploiting the exogeneity resulting from migrant location assignment policy.

3 Research Design

3.1 Data

My data is sourced from the cross-sectional American Community Survey (ACS) microdata, obtained from IPUMS. The geographic level of analysis is the Public Use Microdata Area (PUMA), which is the most granular geographic identifier that is publicly available. PUMAs consist of 100,000 to around 200,000 residents, from which I subset for migrants, defined as non-citizens born outside of the U.S. I also limit my sample to those within the age range of 18 to 65 and exclude those living

in group quarters. I deviate from the standard literature (Card, 2009; Jaeger et al., 2018) that uses the Metropolitan Statistical Areas (MSAs) as their geographic unit of analysis, as PUMAs are smaller than MSAs, and therefore more accurately reflect local network effects. My sample is restricted to the years 2005 to 2019. 2005 is the earliest year where PUMA as a geographic level is included as a geographic variable, while I do not use data after 2019 due to the migration and labor market shocks from the COVID-19 pandemic. However, a limitation to keep in mind with the ACS data is that it may undercount workers without legal status, a group whose network effects are particularly important for their labor outcomes. I restrict the migrants in my sample to be from the top 20 countries with the most migrants in the U.S. This is to ensure that there is sufficient statistical power and geographic variation across migrants from the same origin country.

Because of the decadal change in PUMA boundaries in 2012, I merge the PUMAs before and after the boundary change using the 2000s-to-2010s crosswalk provided by NHGIS. I match the PUMAs one-to-one, assigning each 2000s PUMA the 2010s PUMA identifier that has the highest population match weight with the 2000s PUMA. Because the population of each 2000s PUMA can be distributed unevenly across different 2010s PUMAs, I drop the PUMAs from the 2000s data where the maximum population match is less than 60%. This is to ensure that the PUMAs still stay meaningfully consistent across the decades, without dropping too many PUMAs in my data. I retain about 80% of all PUMAs from 2005 to 2011, before the new PUMA boundaries are used in the 2012 data.

Another limitation of the microdata is the lack of granular information on exact hours worked in a year. My main outcome variable is hourly wages, which is not directly observable in the data. In addition, individual-level hours worked per week are unavailable for data from 2008 to 2018. Instead, I use available data on intervalled hours worked per week, using the median of each interval to estimate the hours that a migrant worked per year. I then use this measure along with data on annual income to construct an estimate for hourly income. This poses limitations to the validity of my data on hourly wage. In light of this, I impose bounds on the hourly wage to be above \$1 and below \$500, based on my prior on implausible hourly wages.

I construct the shift-share instrument using the full 2005-2019 migrant sample, which includes

2,112,504 observations. On the other hand, the regression sample is restricted to new migrants who migrated at year t or $t - 1$. For example, someone who immigrated in 2015 would be included as an observation only in the 2015 and 2016 regressions. Restricting the regression sample to just new migrants is in line with the theorized mechanism here, as information on labor markets is most valuable to migrants right after arrival, when they face high labor market frictions due to unfamiliarity and uncertainty. The final dataset of new migrants from 2010 to 2019 has around 29,000 observations, after dropping observations with incomplete data. The earliest year in the regression sample is 2010, because the instrument is lagged by 5 years, such that 2005 data is used to construct the instrument for 2010. I further explain my data cleaning and variable construction in Appendix A.

3.2 Why not a Linear Model?

Assume the following linear model,

$$Y_{ijkt} = \alpha \text{ethnic}_{jkt} + \beta' X_{it} + \beta_0 + \gamma_j + \mu_k + \lambda_t + \epsilon_{ijkt} , \quad (1)$$

where *ethnic* is the proportion of migrants from origin country k living in PUMA j at year t , and i indexes individual. X is a vector of controls specific to individual i , including race, education, gender, year of immigration, the interaction of education and age, and a quadratic function of age to account for the empirically concave function of wages against age. γ_j, μ_k , and λ_t are fixed effects for PUMA j , origin country k and year t respectively. We are interested in the coefficient on ethnic, α .

α is the conditional association between the ethnic composition of a neighborhood and an immigrant's wage after controlling for individual confounders. However, the major limitation of this approach is that running a regression of immigrant wages on ethnic composition would produce biased estimates of the true effect of ethnic composition on immigrant labor outcomes, compromising the internal validity of this study. This is because of selection: immigrants choose the neighborhoods that they reside in, implying that there is endogeneity in the model arising from simultaneity bias. This means that higher immigrant income in a region attracts more immigrants, which in turn also increases the ethnic composition within that area. When this happens, a change in the outcome

variable also causes a change in the regressors, and the error term will be correlated with the ethnic composition variable. We are therefore unable to establish a causal effect of ethnic network effects on immigrant labor outcomes.

I estimate this linear model with an OLS regression, clustering standard errors at the PUMA level to allow for the non-independence of error terms within the same local labor market, i.e. in a PUMA. I do not find robust, statistically significant results of ethnic shares' effects on new migrants' wages. The OLS estimates of α are larger than the estimates of the instrumented coethnic shares presented later. This suggests that a simple OLS model biases the coefficients upwards. The endogenous selection problem appears here: migrants choose locations where wages are higher, such that locations with higher wages see higher concentrations of migrants. This, however, cannot be interpreted as causal.

Table 1: OLS Estimates of Coethnic Network Share on Wages

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
Network share	0.451	0.495	0.050	0.070*
	(0.944)	(1.02)	(0.031)	(0.036)
p-value	[0.633]	[0.629]	[0.106]	[0.051]
Survey weights	No	Yes	No	Yes
Observations	35,553	35,553	35,553	35,553
R ²	0.21960	0.22262	0.31038	0.32441
Within R ²	0.06470	0.05785	0.11016	0.10251
YEAR fixed effects	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓

Notes: Standard errors clustered at the PUMA level are reported in parentheses. P-values are reported in brackets. Controls include age, age², gender, years since migration, race dummies, education dummies, and education $\times$ age interactions. Weighted columns use ACS person weights. Within R^2 is computed after absorbing year, PUMA, and origin-country fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.3 The Shift-Share Instrument

I estimate the effect of ethnic network sizes on immigrant labor outcomes using a shift-share instrument. A shift-share instrument is analogous to a measure of exposure to an exogenous shock to the predictor variable at a broader, global level. The local unit here is the PUMA, and the global unit is at the national level. The instrument assumes that all else equal, the share of the total inflow of migrants from a given origin country that a local area receives is fixed across time. This assumption is justified by the empirical evidence that migrants are drawn to areas with higher coethnic networks (Bartel, 1989). Figure 1 shows the local concentration of Mexican migrants across time in all PUMAs. The shares do change across time, but in general, we see a pattern of PUMAs with high Mexican concentrations retaining a high Mexican migrant share even after 14 years. Many PUMAs with no Mexican migrants in 2005 see an increased share of Mexican migrants later on, but the shift-share instrument is unable to capture this change as the baseline reference time is set at 2005.

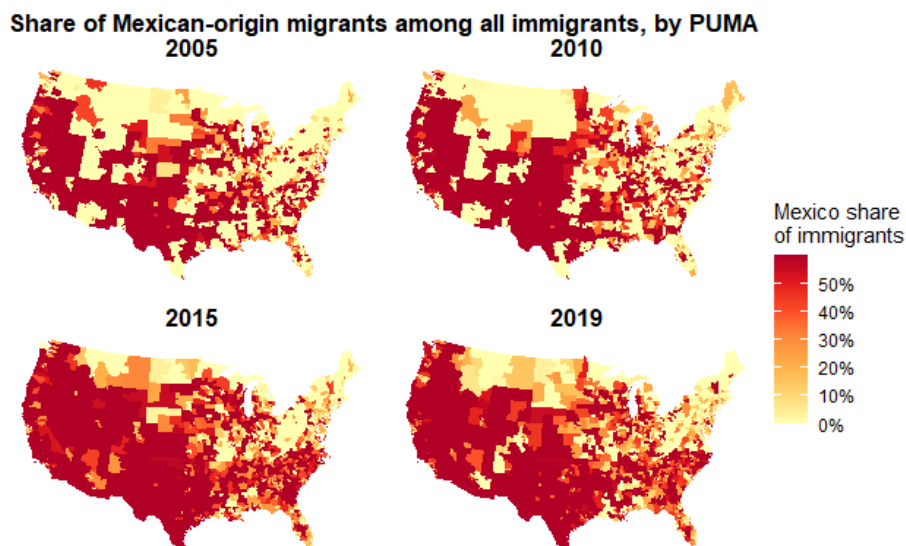


Figure 1: Share of Mexican migrants across PUMAs in different years

Conceptually, the instrument is comprised of two terms: the aggregate *shift* that reflects global changes in levels of the regressor, and a local-level *share* that measures the local unit's exposure to a change in the regressor. The *share* is constructed using a historical reference period. This

is done so that the instrument overall predicts future levels of the endogenous variable based on trends at the historical reference period. Here, my shift-share instrument is constructed by weighing national-level migrant inflow from country k at year t by the share of coethnics from k in PUMA j at baseline year t_0 .

Let M denote the number of migrants. The shift-share instrument used in this study is as follows:

$$Z_{jkt} \equiv \text{share}_{jkt_0} \text{shift}_{kt} ,$$

where the shift is defined as the national inflow of migrants from country k at year t , and the share is the proportion of migrants in PUMA j at year t_0 who are from country k . t_0 here is chosen to be 2005, which is the earliest time period available in my data. For the sake of notation, I assume that changes in M are only driven by inflows, and that outflows are negligible. This is so that the *shift* formulation can be written in terms of changes in the migrant stock. Mathematically, this is formulated as:

$$\text{share}_{jkt_0} = \frac{M_{jkt_0}}{M_{jt_0}} ;$$

$$\text{shift}_{kt} = M_{kt} - M_{k(t-1)} = \Delta M_{kt} .$$

Putting the shift and the share together, we get the shift-share instrument as a predicted exposure to origin-specific migrant inflows, weighted by the baseline importance of origin group k in PUMA j 's immigrant population. We obtain a shift-share instrument for each combination of *origin* k - *PUMA* j - *year* t that are bounded below for statistical power:

$$Z_{jkt} = \underbrace{\frac{M_{jkt_0}}{M_{jt_0}}}_{\text{share}} \underbrace{\Delta M_{kt}}_{\text{shift}} .$$

Empirically, the shift for each *origin* - *year* pair is calculated as the sum of observed new migrants, defined as non-citizens born in k who entered the U.S. at year t . Let y represent the year

of migration. Then, the shift is constructed as:

$$\text{shift}_{kt} = \sum_i \mathbb{I}\{i \mid \text{origin} = k, y = t\} .$$

Note that my instrument differs from the canonical shift-share instrument in immigration literature, such as in Altonji and Card (2001) and Card (2009). The canonical shift-share instrument is used to predict the total number of migrants that will migrate to local area j , summed up over origin countries, indexed by k :

$$Z_{jt} = \sum_k \underbrace{\frac{M_{jkt_0}}{M_{kt_0}}}_{\text{share}} \underbrace{\Delta M_{kt}}_{\text{shift}} .$$

My shift-share instrument differs from this canonical instrument in the formulation of the share. The denominator is traditionally the total number of migrants from country k in the country at time t_0 , which is M_{kt_0} in the instrument ($\frac{M_{jkt_0}}{M_{kt_0}}$). Instead, I use $\frac{M_{jkt_0}}{M_{jt_0}}$. M_{jt_0} is the denominator, which is the total immigrants living in PUMA j at t_0 . This is such that the shares reflect the shares of migrants in a PUMA who are from country k at the historical reference period. The share component here represents how significant the k origin migrants are in a PUMA's migrant population at the baseline period. This is so that the instrument better reflects the concentration of migrant groups within cities, instead of just the arrival of migrants. In my instrument, migrants within a PUMA sum to 1 across origin countries, while migrants from the same origin country in the canonical instrument sum to 1 across PUMAs. The shares here are the measure of exposure to national inflows, and interacting the shares and the shifts in this manner produces a measure of local migrant group concentration. The share measures the historical ethnic composition of a PUMA. The proportion of existing migrant networks interacted with a country-specific migrant inflow shift would then capture the concentration of migrant networks specific to a country k . If the historical share of migrants in a PUMA who are from country k is low, then we expect the instrument to reflect the lower level of predicted migrant shares, absent other shocks.

The shifts work in the same way. If national-level migrant inflow from country k decreases at year t , we would expect less migrants to migrate to any given j than before. The instrument will capture this and reflect a lower coethnic concentration. In short, the instrument provides a prediction of

local coethnic concentration at a given year based on historical ethnic concentration and current coethnic inflow.

In the standard literature, the shift-share instrument is summed up over the local levels to produce an aggregate weighted exposure metric (Borusyak et al., 2025). Here, because I am exploiting variation by origin country k , I do not sum over the origin countries to get a weighted prediction of migrant inflow. Instead, I use the individual components of the typical summed shift-share as my instrument instead. I discuss in detail the empirical construction of my shift-share instrument in Appendix A.

3.3.1 Evaluating the Shift-share Instrument

In order for this instrument to identify a causal effect, it has to be exogenous, relevant and satisfy the exclusion restriction.

Exogeneity:

The exogeneity of the instrument relies on either the exogeneity of the shares or the exogeneity of the shifts (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2025). In the canonical immigration literature utilizing shift-share instruments, it is argued that the shares are exogenous (Altonji and Card, 2001; Card, 2001). This is assuming that when the first migrants from different countries arrived in the U.S., there were no coethnics that draw them to a place (as I assume here). Hence, they exogenously chose their initial place of residence, producing an initial exogenous migrant location choice that we use to predict current migrant inflow patterns. It is also common to argue for the shares to be exogenous (Card, 2009). This is based on the assumption that the national inflow rates from each origin country are exogenous to local labor market conditions (Card, 2001).

With regards to exogenous shares, the historical reference time period is ideally chosen such that barring any shocks, the proportion of migrants from country k that migrate to a specific PUMA would be constant every year. However, I do not have data dating back to initial settlement periods, and so I use data from 2005 as my initial reference period. I rely more on the exogeneity of the shift as the source of exogeneity captured in my instrument. This is because I do not have historical data dating back to initial migrant settlement periods, and also because the shares are constructed using

the total number of migrants already present in a PUMA at 2005, which might not be endogenous.

Relevance:

The instrument is relevant if it has an impact on observed coethnic group size, proxying local network effects:

$$Z_{jkt} \rightarrow ethnic_{jkt} .$$

The instrument is relevant as the historical proportion of migrants $_{jk}$ are correlated with the observed current proportion of migrants $_{jk}$. Empirically, the current size of a coethnic group in an area is empirically shown to be highly correlated with the past number of migrants from the same country (Bartel, 1989; Jaeger, 2007). This is due to the pattern of migrants choosing to live where there is an existing coethnic network. I test this in my data and find in the first-stage diagnostics that the shift-share instrument is very strong, indicating a high level of relevance.

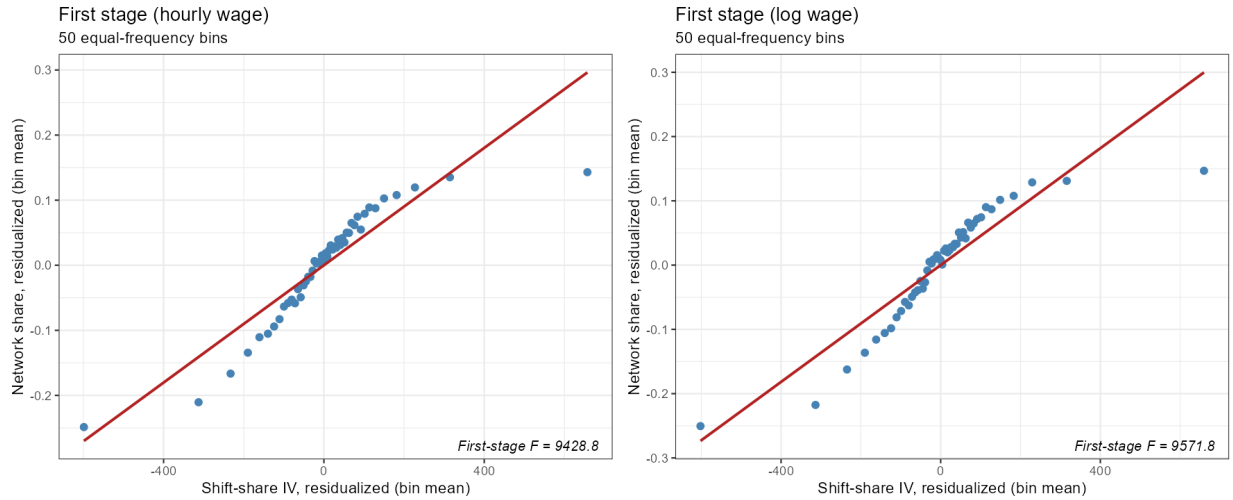


Figure 2: First stage, binned

From the binned scatterplot in figure 2, we see a positive linear relationship between the residualized endogenous variable, coethnic network share, and the residualized instrument. This illustrates that the shift-share is a strong instrument. I use this binned scatterplot because the shift-share instrument produces many values close to 0, leading to many points concentrated at 0

in the unbinned scatterplot. The instrument and network shares are residualized because the first stage specification includes controls and fixed effects. However, by the Frisch-Waugh-Lovell theorem, we can recover the conditional linear relationship between the coethnic share and instrument by regressing the residualized coethnic shares on the residualized instrument.

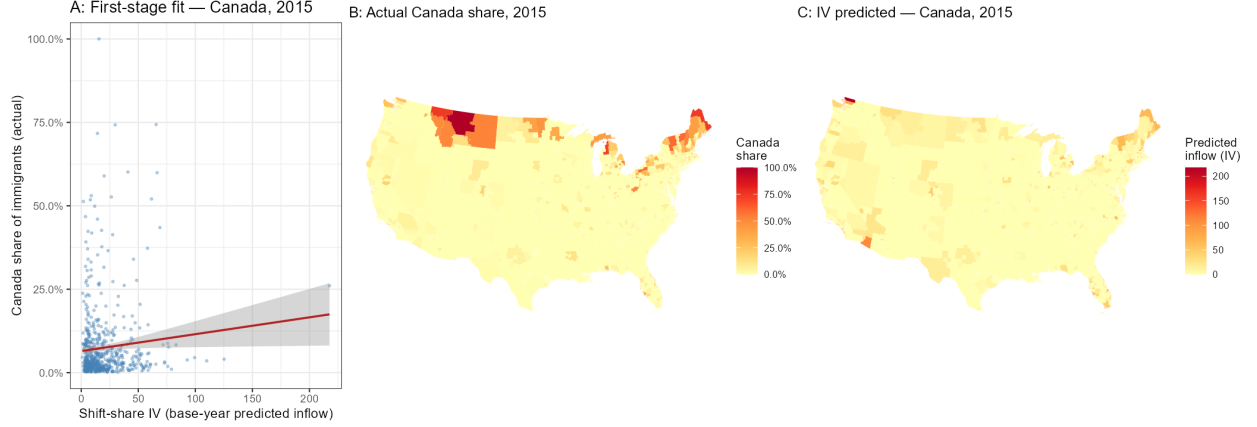


Figure 3: Observed versus instrument-predicted Canadian migrant shares in 2015

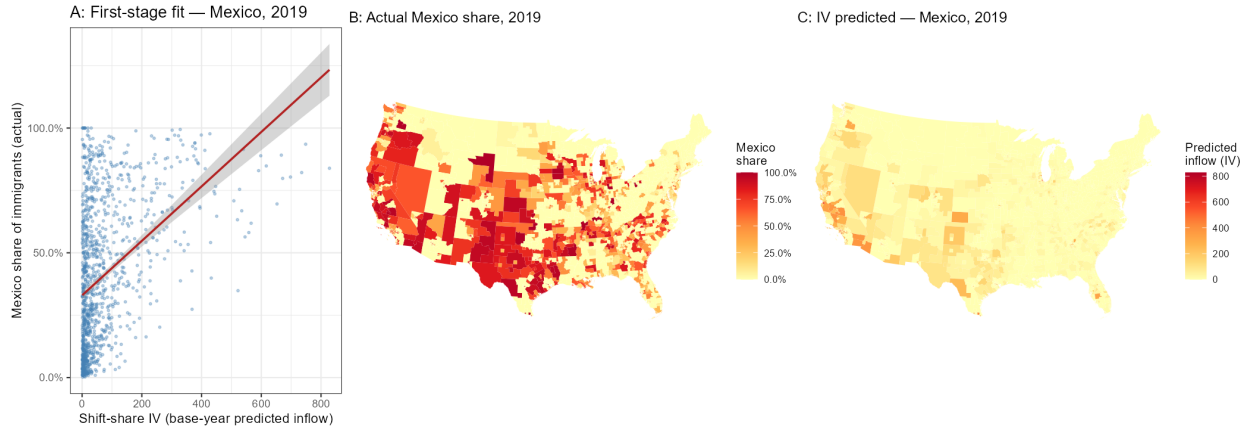


Figure 4: Observed versus instrument-predicted Mexican migrant shares in 2019

Figure 3 maps the instrument’s predicted coethnic networks of Canadian migrants in 2015 and the observed share of Canadian migrants. Though there are differences in the shares across PUMAs, the relationship between the instrument and the observed share is consistent with a positive first stage. Visually, this can be seen especially in the northeast, where the instrument’s spatial patterns correspond to the observed shares. Figure 4 presents the same pattern for Mexican migrants in

2019, and the first stage scatterplot indicates a higher positive relationship between the instrument and observed coethnic shares. These figures provide descriptive visual evidence of the relevance of the instrument.

Exclusion:

The exclusion restriction requires that the instrument affects migrants' labor outcomes only through the endogenous variable, which is the local concentration of coethnics. Visually, this means that conditional on fixed effects and individual-level controls,

$$Z_{jkt} \rightarrow ethnic_{jkt} \rightarrow Y_{ijkt} ,$$

and Z_{jkt} does not directly affect Y_{ijkt} through other channels. Based on my hypothesis that there are network effects on migrant labor outcomes, the exclusion restriction would require that the instrument should affect migrant wages and employment only by affecting the migrant's potential network exposure based on the local coethnic share, and not through other mechanisms.

In this research setting, I assume that the exclusion restriction (of the instrument affecting labor outcomes only through the ethnic concentration) is sufficiently satisfied in both the shift and share components of the instrument. From its construction, the shift component (national-level inflow of migrants) very plausibly reflects shocks from origin countries, and should be independent of local labor market conditions (Card, 2009). This is because the effects of the national shift should be distributed at the local PUMA level such that no single PUMA would be affected directly by a labor supply shock at the national level. However, the shift could reflect national-level labor supply shocks brought about by changing pull factors, such as national visa expansion programs. This could violate the exclusion restriction by creating a labor supply shock that affects wages directly. I address this by using time fixed effects. Time fixed effects absorb changes in local labor market conditions that affect all PUMAs at the same time, which should address such time-specific national shocks. Another potential exclusion violation concern with the shifts is local labor demand shocks for migrants from a specific country. This violates the exclusion restriction, if specific cities attract migrants from certain origins more than others. This is a valid concern, but theoretically, there is no single local labor market that can significantly drive national inflow patterns from a single origin

country. Using this logic combined with the PUMA, year, and origin country fixed effects, I assume that local labor demand shocks does not pose a meaningful threat to the exclusion violation in this case.

The shares should also not affect migrant wages at year t , as the shares are predetermined at year $t_0 = 2005$, and therefore should be independent of wages at a later time. However, there could still be concerns about violations of the exclusion restriction with regards to the shares. For example, migrants could be clustered in certain industries or occupations, which have different wages. I check for this in the heterogeneity analysis, though the interpretations of the coefficients have to be done cautiously as occupations and industries can be post-treatment outcomes. I also introduce PUMA fixed effects, which absorbs time-invariant PUMA characteristics that could be correlated with baseline ethnic shares. This would mitigate the exclusion violation concerns in situations such as migrants clustering in industries.

Additionally, an implication of the exclusion restriction is that it should be independent of other potential confounders that affect wages, conditional on controls and fixed effects. Similar to the previous discussion, the PUMA, year, and origin country fixed effects address this by absorbing time-invariant local labor market characteristics, national shocks common to all migrants in a given year, and persistent differences across migrant-origin groups. Individual controls account for observable wage determinants such as age, gender, education, race, and years since migration. Under this assumption, the remaining variation in the instrument reflects differential exposure to national origin-specific inflows, and not unobserved wage shocks. With these fixed effects, within-PUMA, within-origin, and within-year variation in predicted exposure to country-specific migrant inflow is what allows for identification. This is a weaker assumption than the regular exclusion restriction, but the shift-share instrument I use here fulfills the exclusion restriction, conditional upon unobserved exclusion-violating factors being specific to a given PUMA, year, or origin country. It is important to note that the IV estimates should be interpreted as causal only under the maintained assumption that such PUMA-origin-year wage shocks are not driving the relationship.

Still, there could be channels for exclusion violation within PUMAs, years, and origin countries. These could be internal competition for the same jobs, labor market general equilibrium effects, or

other unobservable factors. We cannot test the exclusion restriction directly, but in addition to the fixed effects and individual-level controls, I run robustness checks on mechanisms affecting migrant employment outcomes. If migrants are competing internally for jobs, then the Beaman (2012) model would demonstrate how different time-specific cohort of migrants compete or help other cohorts. If the instrument affects migrant labor outcomes mainly through labor market equilibrium effects, I will be able to parse out the short- and long-term market adjustment effects through the dynamic bias correction by Jaeger et al. (2018). Though the robustness checks cannot conclusively rule out all mechanisms aside from network effects, they still allow for checks on the primary alternative mechanisms (internal competition and labor market equilibrium adjustments) through which migrant concentration affects new migrants' labor outcomes. In this way, they provide a more convincing narrative of network effects on new migrants' labor outcomes.

3.4 IV Estimation Strategy

The IV model is as follows. Let Y represent hourly wage. I also use the probability of employment as the outcome variable Y to check for network effects on employment outcomes. The 2SLS estimation is as follows.

$$\text{First stage: } ethnic_{jkt} = \pi Z_{jkt} + \beta' X_{ijkt} + \gamma_j + \mu_k + \lambda_t + u_{ijkt} , \quad (2)$$

$$\text{Second stage: } Y_{ijkt} = \alpha \widehat{ethnic}_{jkt} + \beta' X_{ijkt} + \gamma_j + \mu_k + \lambda_t + \epsilon_{ijkt} , \quad (3)$$

where $Cov(\epsilon_{ijkt}, Z_{jkt}) = 0$. $\gamma_j, \mu_k, \lambda_t$ are PUMA, origin country, and year fixed effects respectively. X_{ijkt} are controls on individuals' observable characteristics, i.e. a quadratic function of age, gender, race, education, and age interacted with education. Since year, origin, and PUMA fixed effects were used in addition to observable individual characteristics, the identification in the regression comes from coethnic concentration variation within origin country, within PUMA, and within year.

4 Results

Regressions here are run on a sample of new migrants, defined as migrants who arrived in the U.S. a year, or less, ago. Standard errors are clustered at the PUMA level, to account for systematic

differences in wages across geography.

4.1 Network Effects on New Migrant Wages

The specification used here in table 2 is based on the IV estimation strategy in section 3.4. I use both hourly wage (columns 1-3) and log hourly wage (columns 4-6) as the outcome variables. Population weights from the ACS were used in specifications 3 and 6 to weigh observations. All regressions include gender, a quadratic function of age and years since migration as controls, and education, race, and their interaction are included as additional controls in specifications 3 and 4. The first-stage diagnostics indicate that the shift share instrument is not weak, as the F-statistics are very high, all being above 9000.

For all specifications, there are no significant coefficients on the instrumented coethnic network size, though note that the coefficients here are lower than those in the baseline OLS regression, some even being negative where the coefficients were all positive in the endogenous regression specification. (Columns 2, 3, 5, and 6 correspond to the four columns in the OLS regression table.) This would mean that the OLS estimates are biased upwards, capturing endogeneity whereby migrants sort into places with higher wages. The direction of network effects on wages is negative for specifications 1-4. This is in line with predictions reflecting market equilibrium effects. Increased labor supply, measured through network shares, would decrease wages on a downward-sloping labor demand curve. However, specifications 5 and 6 indicate around a 4% increase in new migrant wages for every 100 percentage point increase in the coethnic share, which would contradict the labor supply prediction.

Therefore, I am unable to reject the null hypothesis that the instrumented ethnic share has no effect on new migrants' wages. The estimates are imprecise. There is no evidence that the effects of networks are small, as the standard errors are large relative to the estimates. The sign of the effects on migrant wages is also ambiguous: the 95% confidence intervals include negative and positive values.

Table 2: Network Effects on Immigrant Wages: Baseline IV Estimates

	wage_hourly			ln_wage		
	(1)	(2)	(3)	(4)	(5)	(6)
Network share	-3.85	-0.571	-0.009	-0.102	0.040	0.041
	(2.85)	(2.56)	(2.81)	(0.090)	(0.076)	(0.087)
95% CI lower	-9.44	-5.59	-5.52	-0.278	-0.109	-0.130
95% CI upper	1.74	4.45	5.50	0.074	0.189	0.212
Educ. & race controls	No	Yes	Yes	No	Yes	Yes
Survey weights	No	No	Yes	No	No	Yes
Observations	28,876	28,876	28,876	28,876	28,876	28,876
F-test (1st stage), network_share	9,571.8	9,428.8	8,997.9	9,571.8	9,428.8	8,997.9
R ²	0.18840	0.21300	0.99360	0.26927	0.30419	0.99449
YEAR fixed effects	✓	✓	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. 95% confidence intervals are calculated as coefficient ± 1.96 times the clustered standard error. All regressions include year, PUMA, and origin country fixed effects. Sample: recently arrived immigrants (years since migration ≤ 1). Network share instrumented by the base-year (2005) shift-share IV. Columns (1)–(3): dependent variable is hourly wage. Columns (4)–(6): dependent variable is log hourly wage. Education and race controls include education dummies, race dummies, and education $\times$ age interactions. Controls included in all specifications are gender, a quadratic function of age, and years since migration (0 or 1).

4.2 Network Effects on New Migrant Employment Outcomes

Table 3 shows the results of an estimated linear probability model of employment, as observed in the data as employed and unemployed new migrants in the labor force. All regressions include gender, a quadratic function of age and years since migration, as well as education, race, and their interaction as controls. Column 1 does not include population weights in the regression, while specification 2

does.¹

The F-statistics for the first stage are very high here again, with the lowest being around 9200. There are no significant effects. Looking at the correlational direction, the magnitudes are positive for the sample, indicating that those who are surrounded by a higher share of coethnics in the sample have a 3 percentage points higher probability of getting a job. However, the coefficient is almost 0 in the regression with population weights. This could mean a barely detectable effect when we generalize to the population, though this is ambiguous as the confidence intervals are wide. Note that the interpretation of the population-weighted estimates should be done carefully for reasons explained in Appendix D.

Table 3: Network Effects on Immigrant Employment: IV Estimates

	emp	
	(1)	(2)
Network share	0.032 (0.030)	0.003 (0.031)
95% CI: lower	-0.027	-0.058
95% CI: upper	0.091	0.064
Educ. & controls	Yes	Yes
Survey weights	No	Yes
Observations	29,657	29,657
F-test (1st stage), network_share	9,327.3	9,206.8
R ²	0.12418	0.99352
YEAR fixed effects	✓	✓
puma_final fixed effects	✓	✓
origin fixed effects	✓	✓

Clustered standard errors at PUMA level in parentheses. 95% confidence intervals are calculated as coefficient ± 1.96 times the clustered standard error. All regressions include year, PUMA, and origin country fixed effects. Controls include gender, a quadratic function of age, years since migration, education, race, and education interacted with age. Sample: recently arrived immigrants (years since migration ≤ 1). Network share instrumented by the base-year (2005) shift-share IV. Dependent variable: employment indicator (1 = employed, 0 = unemployed). Sample includes all new migrants in the labor force.

4.3 Controlling for Total Immigrants

A potential concern in the previous models is that the shift-share instrument violates the exclusion restriction because of its construction as a proportion, when the total number of migrants could

¹Coefficient in the population-weighted estimation is one order of magnitude smaller than the unweighted estimations. Similar patterns occur across different specifications in this paper, whereby the population weights change the estimate and/or the standard errors by a large margin. I discuss the distribution of the population weights in Appendix D.

have scale effects on the wage. Controlling for the total number of immigrants already in a PUMA ensures that identification comes from the variation in the composition of the migrant population, rather than its overall size. In other words, for PUMAs with the same number of total immigrants, are new migrants more likely to have higher wages if the predicted count of coethnics is higher?

Again, we see no significant effects of coethnic numbers on new migrants' wages. The correlational magnitudes are also small, with an increase in hourly wages of around 0.04% if the network share increases by 100 percentage points. Generally, this means there are no effects of the composition of the migrant population on new migrants' wages. The standard errors are again large relative to the estimates, producing wide confidence intervals that contain positive and negative values. Hence, it is unclear what the direction of the effects are, or if the true effects are close to 0.

Table 4: Controlling for Total Immigrant Population in PUMA

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
Network share	-0.566	0.011	0.040	0.041
	(2.54)	(2.82)	(0.076)	(0.087)
95% CI: lower	-5.54	-5.52	-0.109	-0.130
95% CI: upper	4.41	5.54	0.189	0.212
Immigrant pop. (PUMA)	0.0001*	7.45×10^{-5}	-1.66×10^{-7}	-1.51×10^{-6}
	(7.48×10^{-5})	(8.74×10^{-5})	(2.04×10^{-6})	(2.29×10^{-6})
Survey weights	No	Yes	No	Yes
Observations	28,876	28,876	28,876	28,876
F-test (1st stage), network_share	9,428.8	8,999.9	9,428.8	8,999.9
YEAR fixed effects	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. 95% confidence intervals are calculated as coefficient ± 1.96 times the clustered standard error. All regressions include year, PUMA, and origin country fixed effects. Sample: recently arrived immigrants (years since migration ≤ 1). Network share instrumented by the base-year (2005) shift-share IV. All columns include education and race controls. Immigrant population is the total survey-weighted immigrant count in the PUMA-year cell.

5 Heterogeneity Analysis

Given that migrant wages are moderated by different factors, I analyze the heterogeneous effects of network effects by country income group, region, occupation, and industry. In each estimation, I am interested in asking if different groups of migrants rely more (or less) on networks to access higher-paying jobs. I explain the categorization in detail in Appendix B. Appendix C presents the binned scatterplots of the heterogeneous effects by each category.

5.1 Origin Country Income Group

I refer to the World Bank’s GNI-based income classification of countries to categorize migrants’ origin countries as low-income, lower-middle, upper-middle or high-income. I use the World Bank’s classifications from 2010 [World Bank]. I hope that using a year somewhat in the middle of all the years in my sample will better reflect the average mechanisms through which a country’s income classification would affect its expats’ earning potential throughout all years in the sample. This could include the average economic realities on the ground shaping human capital, or the average perception of migrants’ countries across all years in the sample affecting employers’ labor demand. However, this is an assumption that I make, and the year can arbitrarily change a country’s classification. For example, China’s classification changed from lower-middle income to upper-middle income in 2010, and its upper-middle income status is reflected in this analysis. Nevertheless, the general variation between countries should capture the overall differences between countries’ income groups sufficiently enough to reflect general patterns among migrants based on their origin country’s income status. Another issue with country-level classification is that the ACS does not provide granular breakdowns of origin country in certain areas, such as ”Africa” and the ”West Indies” that include many different countries at different income levels. I do my best to capture a region’s income-group based on the ACS and World Bank classification in this case, using the general average of the countries’ income status in the region. Because of this, there are no absolute low-income origin countries in my sample, leaving me with only lower-middle, upper-middle, and high-income countries.

The baseline in this model is upper-middle income country groups. The relationship between a migrant’s wages and coethnic share differs significantly by their origin country’s income status. Migrants from lower-middle income countries benefit more from ethnic networks than those from upper-middle income countries. Looking at the absolute hourly wage regression with population weights (column 2), a migrant from a lower-middle income country earns on average \$15.3 more from a 100 percentage point increase in coethnic share, compared to migrants from upper-middle income countries. Assuming that the regression specification is accurately specified under IV assumptions, we can interpret this as a causal effect, though we are unsure if the mechanism works through information exchange and matching, labor demand, or other factors. This would suggest that

network effects are helpful to migrants from lower-income countries. However, these results are not robust to all specifications, as the coefficient on the sample regression for migrants from lower-middle income countries is not significant.

On the other hand, migrants from high income countries benefit less from their coethnic networks. In fact, the correlation between ethnic network size and the average wage of a migrant from a high-income country relative to an upper-middle income country is negative. The absolute magnitude is even larger here compared to lower-income country moderation effects. I refer to the population-weighted specification in column 2. Compared to migrants from upper-middle income countries, a 10 percentage point increase in coethnics share for migrants from high income countries is associated with a decrease of \$4 in hourly wage. This implies that the local coethnic share is more negatively associated with migrant wages for migrants from high-income countries than migrants from upper-middle income countries. The mechanism here is unclear as well. It is possible that this follows from labor supply theory: those from high-income countries compete with each other for better-paying jobs, depressing wages within the same ethnic group in the same area. Alternatively, those from high-income countries have sufficient resources to access good job opportunities on their own and do not rely on network effects, and those who do rely on network effects are only able to access lower-paying jobs. We are unable to discern the exact mechanism here, but under IV assumptions, this would imply a causal moderating effect of origin countries' income status on network effects.

Table 5: Heterogeneity by Origin Country Income Group

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
High-income origin $\times$ network share	-31.2**	-39.9**	-1.33***	-1.11***
	(14.6)	(18.3)	(0.363)	(0.364)
Lower-middle-income origin $\times$ network share	10.1	15.3**	0.431***	0.521***
	(6.56)	(6.11)	(0.144)	(0.160)
Survey weights	No	Yes	No	Yes
Observations	28,826	28,826	28,826	28,826
First-stage F: high-income origin	3,180.9	2,951.2	3,180.9	2,951.2
First-stage F: lower-middle-income origin	3,696.8	2,999.6	3,696.8	2,999.6
Year fixed effects	✓	✓	✓	✓
PUMA fixed effects	✓	✓	✓	✓
Origin-country fixed effects	✓	✓	✓	✓

Clustered standard errors at the PUMA level are reported in parentheses. All regressions include year, PUMA, and origin-country fixed effects. The sample is restricted to recently arrived immigrants, defined as migrants with years since migration ≤ 1 . Network share is instrumented by the base-year 2005 shift-share IV. The reference category is upper-middle-income origin countries, based on the World Bank 2010 classification. All columns include education, race, and immigrant-population controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 shows the results of a joint Wald test testing the null hypothesis that the coefficients on all the instrumented network shares interacted with both income groups are equal to 0. The highest p-value on the joint tests is 0.055, which would mean that we can reject the null hypothesis that there are no significant interactions by origin country income groups. If we use a Bonferroni correction with $\alpha = 0.05$, such that $\alpha_B = 0.0125$, we still reject the null hypothesis in two specifications. The significance is not robust while controlling for the family-wise error rate with the Bonferroni correction, but it does suggest that there is strong evidence that the network effects of migrants' wages are moderated somewhat by the income status of the countries they are from.

Table 6: Joint Wald test: income group heterogeneity

Spec.	Outcome	Weights	F-statistic	Restrictions	p-value
(1)	Hourly wage	No	2.897*	2	0.055
(2)	Hourly wage	Yes	4.333**	2	0.013
(3)	Log wage	No	9.679***	2	0.000
(4)	Log wage	Yes	7.871***	2	0.000

Cluster-robust F -test (clustered at PUMA level). H_0 : all network share $\times$ group interaction coefficients jointly zero.

Restrictions = number of groups tested (reference category excluded). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Region

Similar to income groups, we can check if the geographic regions of migrants' origin countries moderate network effects on wages. Using Canadian migrants as the baseline reference group, European migrants are significantly likely to earn less in response to an increase in coethnic concentration. There is also evidence that African migrants tend to earn less than Canadian migrants in response to the same increase in local coethnic share, and Asian migrants earn more, though these effects for African and Asian migrants are not robust to all estimations. According to the network effects model and IV assumptions, Asian migrants are plausibly relying more on their networks to secure higher-paying jobs, and European and African migrants less. The estimates suggest that Latin American migrants are also more likely to earn slightly less living in an area with a higher coethnic share, but the results are not statistically significant.

The joint Wald tests on the four regressions are significant (table 8), with the p-values on all tests being lower than $\alpha = 0.05$. Similar to the previous section, we can also use a Bonferroni correction with $\alpha_B = 0.0125$, correcting for the four tests. Two specifications would then be significant, though the two other p-values would still be very low despite failing the Bonferroni adjustment test. This suggests that there are possible differences in how much network effects affect migrant wages based on where the geographic regions that migrants are from.

Table 7: Heterogeneity by Region of Origin

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
region::Africa:network_share	-14.1 (13.4)	-17.9 (17.5)	-0.902* (0.488)	-1.34* (0.718)
region::Asia:network_share	6.19 (5.06)	9.32 (5.92)	0.250* (0.138)	0.318* (0.174)
region::Europe:network_share	-25.4** (10.6)	-37.0*** (13.4)	-1.35*** (0.405)	-1.68*** (0.484)
region::LatAm:network_share	-4.79* (2.50)	-4.37 (3.78)	-0.026 (0.110)	0.002 (0.129)
Survey weights	No	Yes	No	Yes
Observations	28,826	28,826	28,826	28,826
F-test (1st stage), region	875.71	686.43	875.71	686.43
F-test (1st stage), region	2,547.5	2,203.5	2,547.5	2,203.5
F-test (1st stage), region	3,006.6	2,469.5	3,006.6	2,469.5
F-test (1st stage), region	1,568.9	1,576.3	1,568.9	1,576.3
YEAR fixed effects	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. All regressions include year, PUMA, and origin country fixed effects.

Sample: recently arrived immigrants (years since migration ≤ 1). Network share instrumented by the base-year (2005)

shift-share IV. Reference: Canada. All columns include education, race, and immigrant population controls.

Table 8: Joint Wald Test for Region-of-Origin Heterogeneity

Spec.	Outcome	Weights	F-statistic	Restrictions	p-value
(1)	Hourly wage	No	2.951**	4	0.019
(2)	Hourly wage	Yes	3.184**	4	0.013
(3)	Log wage	No	4.430***	4	0.001
(4)	Log wage	Yes	4.880***	4	0.001

Notes: The table reports cluster-robust F -tests clustered at the PUMA level. The null hypothesis is that all network share $\times$ region-of-origin interaction coefficients are jointly zero. Restrictions refer to the number of region-of-origin groups tested, excluding the reference category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Occupation

We can also hypothesize that migrants within different occupations rely on networks to different extents. The baseline occupation category here is the unclassified group in the ACS occupation variable.

Table 9 presents the results of the heterogeneity analysis by occupation. Those in management, business, science and arts (one category) as well as the military earn more in PUMAs with larger coethnic networks. There could be an unseen endogeneity concern, as higher-earning occupations require higher skills, and vice versa. However, education controls were used in the regression. The education controls for each migrant should proxy migrants' skill levels, and isolate the variation within educational attainment. If this is the case, then we can argue that the heterogeneity by occupation does reflect differences in how network effects benefit certain occupations more than others. A potential mechanism here could be that more prestigious jobs, such as those in management, business, science, and arts, are more competitive in the general labor market, and hence network effects are favorable to those who are able to take advantage of referrals. It is important to note here that the regression coefficient for military should be interpreted carefully, as the category only has around 80 observations. This means that the estimate is likely to be biased, and should be considered cautiously.

On the other hand, those in natural resources, construction and maintenance (one category),

production, sales/office, as well as service earn less in PUMAs with a higher coethnic concentration. A potential mechanism could be that migrants from the same country are substitutes for each other in these occupations, and so they are competing with each other for the same occupations. A higher coethnic share here would drive down wages.

It is likely that occupations are a post-treatment outcome of network exposure, not a pre-treatment characteristic of migrants. This is true if information sharing is the main mechanism through which coethnic concentration affects wages. Referrals to job opportunities is a sorting mechanism, as the sharing of information affects what occupations one is exposed to. If it is true that occupations are a post-treatment outcome, then we cannot claim causal network effects on migrants' labor outcomes as differing based on migrants' occupations. This would merely illuminate what industry are likely to be exposed to network effects or internal competition instead.

I check with a joint Wald test if any coefficient on the instrumented occupation-heterogeneous network shares are significant. From table 10, the tests confidently reject the null hypothesis that none of the coefficients are zero. They also pass the Bonferroni-adjusted joint test. Hence, coethnic network effects on wages do differ based on the occupation of a migrant, but the results cannot be interpreted as an occupation-moderating effect on wages.

Table 9: Heterogeneity by Occupation

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
Management, business, science, and arts	17.3*** (3.23)	20.4*** (3.56)	0.733*** (0.086)	0.820*** (0.093)
Military	2.75 (38.0)	26.5 (31.1)	2.24** (1.01)	3.30*** (0.497)
Natural resources, construction, and maintenance	-10.6*** (2.29)	-8.13** (3.66)	-0.210** (0.084)	-0.166 (0.112)
Production, transportation, and material moving	-15.6*** (2.45)	-15.9*** (3.85)	-0.565*** (0.078)	-0.538*** (0.096)
Sales and office	-7.84*** (2.83)	-5.92* (3.08)	-0.280*** (0.107)	-0.232* (0.121)
Service	-13.1*** (2.28)	-10.6*** (2.27)	-0.477*** (0.096)	-0.412*** (0.096)
Survey weights	No	Yes	No	Yes
Observations	28,876	28,876	28,876	28,876
First-stage F: management/business/arts	2,569.0	2,354.8	2,569.0	2,354.8
First-stage F: military	10,008.9	23,758.3	10,008.9	23,758.3
First-stage F: natural resources/construction	1,959.7	1,920.0	1,959.7	1,920.0
First-stage F: production/transportation	2,312.7	2,301.7	2,312.7	2,301.7
First-stage F: sales/office	2,963.0	2,901.4	2,963.0	2,901.4
First-stage F: service	2,524.2	2,479.2	2,524.2	2,479.2
YEAR fixed effects	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓

Notes: Clustered standard errors at the PUMA level are reported in parentheses. All regressions include year, PUMA, and origin-country fixed effects. The sample is restricted to recently arrived immigrants, defined as those with years since migration ≤ 1 . Network share is instrumented using the base-year 2005 shift-share IV. The reference category is unclassified occupation. All columns include education, race, and immigrant-population controls. First-stage F-statistics correspond to the excluded instruments for each occupation interaction.

Table 10: Joint Wald Test for Occupation Heterogeneity

Spec.	Outcome	Weights	F-statistic	Restrictions	p-value
(1)	Hourly wage	No	30.554***	6	0.000
(2)	Hourly wage	Yes	27.313***	6	0.000
(3)	Log wage	No	40.963***	6	0.000
(4)	Log wage	Yes	43.668***	6	0.000

Notes: The table reports cluster-robust F -tests clustered at the PUMA level. The null hypothesis is that all network share $\times$ occupation-group interaction coefficients are jointly zero. Restrictions refer to the number of occupation groups tested, excluding the reference category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Industry

Similar to occupations, we can check if industries moderate the effects of network on a migrant's earnings. The baseline used in this regression are industries that do not fit into major buckets, categorized as "other." Industry heterogeneous effects are similar to those of occupations. Focusing on statistically significant coefficients, we see that those in finance, insurance, real estate (one category), information, and professional administrative service outearn the baseline with the same coethnic network size. The effects are most pronounced for those in the information industry, such as IT. They outearn the baseline by around \$30 (hourly wage) for every 100 percentage point increase in their network size. More intuitively, a 10 percentage point increase in the local coethnic share is associated with a \$3 increase in hourly wage, relative to the 'other' industry category. On the other hand, those in food services, accommodation, arts and entertainment (one category) earn less than the baseline. This would align with the hypothesis put forth in the earlier section about certain occupations or industries being more competitive than others. There could be a causal mechanism linking network sizes to increased earnings in more competitive industries and occupations, but we are unable to check that here directly. This is because similar to occupations, industry is a post-treatment outcome and can be directly affected by the information shared within networks. We should also interpret this not as a causal moderating effect of industry on network effects, but rather as a descriptive, exploratory finding.

Table 11: Heterogeneity by Industry, First-Stage Diagnostics

	wage_hourly		ln_wage	
	(1)	(2)	(3)	(4)
Agriculture, fishing, and mining	2,007.7	1,952.4	2,007.7	2,030.0
Arts, accommodation, and food services	1,322.7	1,358.8	1,320.1	1,404.9
Construction	877.44	854.41	877.53	874.27
Education, health, and social assistance	1,418.1	1,373.5	1,415.7	1,391.3
Finance, real estate, and rental	1,838.5	1,714.6	1,838.2	1,735.2
Information	2,288.3	2,133.7	2,288.4	2,227.1
Manufacturing	1,040.2	1,019.5	1,039.8	1,044.3
Military	6,405.3	11,231.6	6,399.5	11,318.6
Professional and administrative services	1,527.6	1,340.6	1,527.3	1,366.7
Public administration	1,850.6	1,817.7	1,848.0	1,814.2
Retail trade	1,569.2	1,475.9	1,565.8	1,523.3
Transportation, warehousing, and utilities	1,361.0	1,360.1	1,360.6	1,366.6
Wholesale trade	1,614.7	1,663.1	1,614.2	1,709.0
YEAR fixed effects	✓	✓	✓	✓
PUMA fixed effects	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓

Looking at the joint Wald test in the table below, we can reject the null hypothesis that migrants in all industries experience the same instrumented network effects on wages. This is true also after adjusting the α threshold for the Bonferroni correction. However, we are unable to determine causal moderating effects of industry on network effects.

Table 12: Joint Wald Test for Industry Heterogeneity

Spec.	Outcome	Weights	F-statistic	Restrictions	p-value
(1)	Hourly wage	No	8.566***	13	0.000
(2)	Hourly wage	Yes	5.029***	13	0.000
(3)	Log wage	No	13.459***	13	0.000
(4)	Log wage	Yes	8.720***	13	0.000

Notes: The table reports cluster-robust F -tests clustered at the PUMA level. The null hypothesis is that all network share $\times$ industry-group interaction coefficients are jointly zero. Restrictions refer to the number of industry groups tested, excluding the reference category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Robustness Checks

6.1 Lower bounds on network shares

To conduct a sensitivity test on the results of network effects on migrant wages, I rerun the baseline regressions from Sections 4.1 and 4.2. Here, I use a lower bound on the network shares and lagged shares to keep just more than half the observations. The lower bound on the network share imposed here is 0.05, and the lower bound on the lagged shares is 0.0005. This will filter out the noise coming from *origin-PUMA-year* shares that are very small.

Table 13: Robustness: Lower Bound Sample

	wage_hourly		ln_wage		emp	
	(1)	(2)	(3)	(4)	(5)	(6)
Network share	-0.882	-0.862	0.040	0.036	0.023	0.023
	(3.41)	(3.75)	(0.104)	(0.111)	(0.030)	(0.029)
Survey weights	No	Yes	No	Yes	No	Yes
Observations	20,780	20,780	20,780	20,780	18,719	18,719
F-test (1st stage), network_share	5,546.6	5,440.4	5,546.6	5,440.4	4,738.4	4,974.7
R ²	0.21672	0.99394	0.31258	0.99474	0.08590	0.99303
YEAR fixed effects	✓	✓	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. All regressions include year, PUMA, and origin country fixed effects. Sample: recently arrived immigrants (years since migration ≤ 1). Network share instrumented by the base-year (2005) shift-share IV. Lower bound sample: network share > 0.05 and lagged share $> 5 \times 10^{-4}$. Columns (1)–(4): wage outcomes; all include education and race controls. Columns (5)–(6): employment outcome.

From the wage and employment probability regressions, we do not find a significant effect of instrumented ethnic shares on new migrants' wages or employment. This is in line with results from

Section 4: the lack of significant correlations are confirmed by this robustness check. Standard errors are similarly large relative to the point estimates, which means that we are unable to conclusively rule out effects of larger magnitudes.

6.2 Beaman’s Dynamic Model Specification

Beaman (2012) uses a dynamic model to parse out the different effects that different cohorts of migrants have on each other’s wages. Beaman suggests that migrants who arrive a year earlier (at $t - 1$) still face frictions in the labor market, and hence they compete with migrants arriving at year t . Instead of using the absolute size of the existing networks in a region, she separates out the inflows from the past four years by year. She finds that the coethnic share who arrived a year earlier (at $t - 1$) has negative effects on migrants, supporting her theory that new migrants, even those who arrived a year ago, compete with new migrants (arriving at year t) for the same jobs. Beyond the first year, the number of migrants who arrive two years or more prior to a new migrant have positive effects on new migrants’ employment (Beaman, 2012). This indicates that migrants do take time to assimilate and overcome local labor market frictions, such that they are able to act as information providers for new migrants in their network after more than a year in the local labor market.

Beaman (2012) does not instrument for the network size, but here I adjust her model to instrument the network size, still separating out the network sizes of each cohort up to four years prior to a migrant’s arrival. The new endogenous variables are now the coethnic share in the total migrant population who arrive a certain number of years prior to a migrant’s arrival. In the regression, *YSM* indicates year since migration. Note that *YSM* 1 in the regression table refers to the cohort of migrants who arrive in the same year for an individual in the sample, and *YSM* 5 refers to the cohort size of migrants who arrived 4 years prior.

My regression specification is as follows:

$$\text{First stage: } \mathit{ethnic}_{jkt, YSM=t} = \gamma_{t1} Z_{jkt, YSM=t} + \gamma_{01} + u_{jkt} ,$$

$$\mathit{ethnic}_{jkt, YSM=t-1} = \gamma_{t2} Z_{jkt, YSM=t-1} + \gamma_{02} + u_{jkt} ,$$

$$\mathit{ethnic}_{jkt, YSM=t-2} = \gamma_{t3} Z_{jkt, YSM=t-2} + \gamma_{03} + u_{jkt} ,$$

$$\mathit{ethnic}_{jkt, YSM=t-3} = \gamma_{t4} Z_{jkt, YSM=t-3} + \gamma_{04} + u_{jkt} ,$$

$$\mathit{ethnic}_{jkt, YSM=t-4} = \gamma_{t5} Z_{jkt, YSM=t-4} + \gamma_{05} + u_{jkt} ,$$

$$\begin{aligned} \text{Second stage: } Y_{ijkt} = & \alpha_1 \widehat{\mathit{ethnic}}_{jk, YSM=t} + \alpha_2 \widehat{\mathit{ethnic}}_{jk, YSM=t-1} + \alpha_3 \widehat{\mathit{ethnic}}_{jk, YSM=t-2} \\ & + \alpha_4 \widehat{\mathit{ethnic}}_{jk, YSM=t-3} + \alpha_5 \widehat{\mathit{ethnic}}_{jk, YSM=t-4} + \beta' X_{ijkt} + \beta_0 \\ & + \gamma_j + \mu_k + \lambda_t + \epsilon_{ijkt} . \end{aligned}$$

Following this specification, there are no statistically significant effects of a given migrant's cohort on those who arrive after them. We can however look at the pure directional effects to learn the direction of the correlations.

6.2.1 Results: Employment Probability

The number of a migrant's coethnics who arrive in the same year has a negative correlation with their linear probability of employment (refer to table 14). This supports predictions from the traditional labor market model of decreased wages when labor supply increases. However, unlike Beaman (2012), I do not find that the negative effects extend to the previous cohort of migrants who arrived at $t - 1$. There could be a quadratic relationship between the cohort size of migrants who arrived prior to a new migrant and new migrants' employment. This is because coefficients on employment are positive for the cohort size of migrants who arrive one and two years prior, but negative for migrants who arrived three and four years prior. This would mean that the number of longer-term migrants are negatively correlated with a new migrant's wages, and the number of migrants who arrive less than two years prior is positively correlated with a new migrant's wage. If there are causal effects here, it could be because migrants who arrive earlier are less sympathetic to new migrants than more recent migrants, though this cannot be established directly here. This differs from Beaman's (2012) findings that only the number of migrants who arrive in the same and

the previous year are negatively correlated with new migrants' employment, and the cohort size of older migrants have positive effects.

6.2.2 Results: Wages

The results on wages here also differ from Beaman's (2012) findings on wages, though they line up with her theoretical predictions more than the previous results on employment probability. Beaman finds that wages are increasing in the number of migrants who arrive earlier, but here, I see negative effects of migrant cohort numbers on new migrants' wages up to the cohort who migrated two years prior. Differing from the results on wages, coethnics who arrive three and four years prior have a positive correlation with new migrants' wages. These results agree with Beaman's model that predicts positive network effects of migrants who migrated earlier, but negative network effects of migrants who migrated more recently. However, these are not significant and their interpretation should be done cautiously.

Table 14: Robustness: Cohort Decomposition (Beaman 2012)

	wage_hourly		ln_wage		emp	
	(1)	(2)	(3)	(4)	(5)	(6)
YSM 1 (competition)	-236.7 (501.0)	-148.1 (366.5)	-5.34 (12.9)	-5.34 (12.9)	-2.22 (6.60)	-1.42 (25.6)
YSM 2	-140.2 (574.8)	-152.3 (425.5)	-0.403 (14.9)	-0.403 (14.9)	4.34 (6.83)	13.2 (37.1)
YSM 3	-533.4 (714.2)	-468.5 (598.7)	-15.6 (20.5)	-15.6 (20.5)	6.75 (7.61)	24.3 (66.3)
YSM 4	452.1 (588.7)	302.7 (435.9)	11.5 (15.0)	11.5 (15.0)	-2.36 (4.77)	-0.410 (16.9)
YSM 5 (established)	523.6 (506.9)	518.6 (636.5)	11.7 (14.6)	11.7 (14.6)	-7.79 (8.67)	-40.0 (112.6)
Outcome	Hourly wage	Hourly wage	Log wage	Log wage	Employed	Employed
Survey weights	No	Yes	No	Yes	No	Yes
Observations	68,318	68,318	68,318	68,318	69,125	69,125
F-test (1st stage), share_ysm1	57.209	36.086	57.209	57.209	61.408	39.911
F-test (1st stage), share_ysm2	307.15	269.49	307.15	307.15	318.55	282.00
F-test (1st stage), share_ysm3	466.76	400.69	466.76	466.76	483.97	431.18
F-test (1st stage), share_ysm4	535.85	483.93	535.85	535.85	532.25	476.32
F-test (1st stage), share_ysm5	347.19	333.63	347.19	347.19	359.05	343.24
YEAR fixed effects	✓	✓	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. All regressions include year, PUMA, and origin country fixed effects. Sample: recently arrived immigrants (years since migration ≤ 1). Five endogenous variables (coethnic share by years-since-migration cohort) instrumented by five cohort-specific base-year shift-share IVs following Beaman (2012). YSM 1 captures recent arrivals (competition effect); YSM 2-5 capture established coethnics (network effect). *ivf*: first-stage F per endogenous variable; *ivf1*: minimum first-stage F across all five endogenous variables.

6.3 Jaeger et al.’s Dynamic Bias Adjustment

Jaeger et al. (2018) argue that the shift-share instrument conflates both short-term partial equilibrium effects of recent migrant inflow and the long-term effects of the labor market’s adjustment back to general equilibrium in response to past migrant inflow (Jaeger et al., 2018). In other words, the shift-share instrument does not distinguish between the local labor market’s lagged response to migrants who migrated a long time ago, and the short-term impact of increased labor supply from recent migrants. These mechanisms act in opposite directions. In response to increased migrants, the short-term partial equilibrium effects push down migrant wages, and the long-term equilibrium sees increased wages after the markets fully adjust for the short-term drop in wages resulting from the shock. This leads to biased estimates if we do not consider what Jaeger et al. call the dynamic adjustment bias. Following their paper, we can check if the previous results change by introducing a secondary dynamic shift-share instrument that is lagged by 10 years, capturing the long-term adjustment to labor supply shocks, in addition to the short-term instrument already present in my analysis. This instrument would then parse out the long-term market general equilibrium adjustment, and we will have captured the effects of the short-term migrant labor supply shock on the original instrument. This is similar to Beaman’s model, differing only in its lag and mechanism (Jaeger et al.’s conflation of short and long-term market adjustment and Beaman’s network effects).

In order to keep with the proposed 10-year timeline, I use only observations from 2015-2019 in my regression sample, and construct the lagged 10-year instrument from the observations from 2005 onwards. The model is estimated as follows:

$$\text{First stage: } ethnic_{jkt} = \gamma_{11} Z_{jkt} + \gamma_{12} Z_{jk,t-10} + \gamma_{01} + u_{jkt} ,$$

$$ethnic_{jk,t-10} = \gamma_{21} Z_{jkt} + \gamma_{22} Z_{jk,t-10} + \gamma_{02} + v_{jkt} ,$$

$$\text{Second stage: } Y_{ijkt} = \alpha_1 \widehat{ethnic}_{jkt} + \alpha_2 \widehat{ethnic}_{jk,t-10} + \beta' X_{ijkt} + \beta_0 + \gamma_j + \mu_k + \lambda_t + \epsilon_{ijkt} .$$

6.3.1 Results: Employment

From columns 5 and 6 in table 16, both the short and long-term effects of the instrumented coethnic network shares on new migrant’s probabilities of being employed are ambiguous. The sample

regression predicts a negative effect of long-term network size on new migrants' employment, but a positive effect from the short-term network size. However, the signs are flipped in exactly opposite directions in the regression weighted by the ACS population weights. The results are ambiguous and we are unable to draw any meaningful insight on the short- and long-term effects of ethnic networks on new migrants' employment probabilities.

6.3.2 Results: Wages

Results here do not provide statistically significant evidence of either the short- or long-term effects of ethnic concentration on migrant wages. Looking at the directional correlation, the signs on the coefficients in the wage regressions align with Jaeger et al.'s theoretical predictions. The current instrumented network shares are negatively correlated with new migrants' wages. This is consistent with the short-term market partial equilibrium where wages decrease in response to a positive labor supply shock. The long-term, 10-year instrumented network share is positively correlated with new migrants' wages. This is in turn consistent with the positive effects of past supply shocks on current migrant wages, because of the markets adjusting back to general equilibrium, and also through channels such as network effects. These results suggest that the short- and long-term effects of migrant concentrations do affect wages as Jaeger et al. (2018) predict, but they should be interpreted carefully due to their lack of statistical significance.

Table 15: Robustness: Dynamic Adjustment Bias (Jaeger et al. 2018)

	wage_hourly		ln_wage		emp	
	(1)	(2)	(3)	(4)	(5)	(6)
Network share (current)	-41.7 (32.0)	-40.1 (34.4)	-0.338 (0.996)	-0.337 (1.19)	0.318 (0.317)	-0.150 (0.328)
Network share (10-yr lag)	41.8 (31.6)	39.9 (33.2)	0.366 (0.996)	0.383 (1.17)	-0.327 (0.322)	0.112 (0.329)
Outcome	Hourly wage	Hourly wage	Log wage	Log wage	Employed	Employed
Survey weights	No	Yes	No	Yes	No	Yes
Observations	16,468	16,468	16,468	16,468	16,585	16,585
F-test (1st stage), network_share	4,908.6	4,841.3	4,908.6	4,841.3	5,191.5	5,132.0
F-test (1st stage), network_share_lag10	5,204.6	5,228.7	5,204.6	5,228.7	5,339.8	5,334.3
YEAR fixed effects	✓	✓	✓	✓	✓	✓
puma_final fixed effects	✓	✓	✓	✓	✓	✓
origin fixed effects	✓	✓	✓	✓	✓	✓

Clustered standard errors at PUMA level in parentheses. All regressions include year, PUMA, and origin country fixed effects. Sample: recently arrived immigrants (years since migration ≤ 1), 2015–2019 only. Two endogenous variables instrumented by base-year (2005) shift-share IV and the 10-year lagged shift-share IV following Jaeger et al. (2018). *ivf*: first-stage F per endogenous variable; *ivf1*: minimum first-stage F across both endogenous variables.

7 Limitations: Considerations of Validity

7.1 Construct Validity

This study uses the shift-share instrument to instrument for the amount of country-specific migrants in a PUMA, which proxies network effects. We do not directly observe individual interactions, which would be the most relevant measure of network effects. We are instead approximating local network effects by how much a local share of country-specific migrants theoretically change based on aggregate national trends, which might not directly reflect network effects and interactions on the ground. However, the number of coethnics in the same area is an approximate measure for networks and interactions, as is standard in the literature that assume that immigration group sizes affect employment outcomes through the channel of interactions and networks (Bertrand et al., 2000; Beaman, 2012; Patacchini and Zenou, 2012; Battisti et al., 2022).

In addition, another issue with the predicted number of migrants proxying network effects is the scale of the geographic unit. The smallest spatial unit for which ACS microdata is publicly available is the PUMA. This poses a challenge to the construct validity of network effects as well. We expect network effects to be local, but geographic boundaries are often arbitrarily determined, and so we are unable to trace the exact local level of interactions at an accurate level of granularity.

Another limitation of the data is the estimate of hourly wage, which is not directly observed but instead inferred from annual income, divided by the product of usual weekly hours worked and the median of intervalled hours worked per week. We end up with only the best estimate of the hourly wage from the data, so there is more noise than there would be if we had observed data.

7.2 Internal Validity

The internal validity of this study depends mainly on the validity of the shift-share instrument. For the IV estimates to be interpreted causally, the instrument must be relevant, conditionally exogenous, and satisfy the exclusion restriction. The relevance condition is empirically supported by the first-stage results. However, exogeneity and exclusion cannot be tested directly. These assumptions require that, conditional on individual controls and PUMA, origin-country, and year

fixed effects, predicted coethnic networks affect new migrants' wages and employment only through actual local coethnic concentration.

The main threat to internal validity is that the instrument may be correlated with other unobserved PUMA-origin-year factors that independently affect labor outcomes. For example, national inflows from a specific origin country may coincide with labor-demand shocks in industries or cities where migrants from that origin were already concentrated. In that case, the instrument would partly capture local labor-demand conditions rather than only predicted coethnic exposure. Similarly, baseline coethnic shares may reflect persistent local characteristics, such as industry composition, enclave institutions, housing costs, or migrant skill composition, that also affect wages. PUMA, origin-country, and year fixed effects help address time-invariant local factors, persistent origin-group differences, and aggregate year shocks, but they do not fully rule out origin-specific local shocks.

The alternative specifications and robustness checks using different sample restrictions and dynamic models were aimed to improve the internal validity of my results. However, the robustness checks cannot prove the exclusion restriction or eliminate all threats to internal validity. Therefore, the results should be interpreted as causal only under the IV assumptions in section 3.3.1, and the estimates should be understood as the effect of instrumented local coethnic concentration, not evidence of a pure information-network mechanism.

7.3 External Validity

A goal of this study was to contribute externally valid results that other papers utilizing quasi-experimental situations could not generalize to. If results are internally valid, I believe that they would also be externally valid, as the ACS data is representative of the immigrant population, and there is no context-specific immigration factors or other issues affecting wages only in this context.

8 Conclusion

In this paper, I used large-scale ACS data to estimate how network effects affect new migrants' wages. Using ACS data allows me to address external validity concerns from previous studies

that use refugee populations to estimate network effects on migrant wages. However, the main endogeneity concern here is that migrants sort into geographic regions. I use a shift-share instrument that proxies the number of migrants in a PUMA to proxy for network effects. The direction and network effects on migrant wages and employment probabilities are inconclusive. We have evidence that there are large heterogeneous effects of network size on migrant wages by industry, occupation, geographic region of origin and income status of origin country. The implications of these are important for migrants: referrals and information sharing through word of mouth in networks are more important in prestigious and skilled industries such as information. They are also likely to be more important for migrants from lower-middle income countries than for migrants from high-income or upper-middle income countries. European migrants are less likely to benefit from coethnic networks as well. Through the robustness checks using dynamic models proposed by Beaman (2012) and Jaeger et al. (2018), we do not find significant results, but there are indications that the short- and long-term effects of coethnic networks do affect wages and employment through different market adjustment mechanisms.

A Data Construction

Microdata was sourced from the ACS via IPUMS from the years 2005 –2019. I source data only on non-citizens between the ages of 18 to 65 who are born outside of the U.S. and do not live in group quarters.

A.1 Geographic Harmonisation

The geographic unit of analysis is the Public Use Microdata Area (PUMA). PUMAs present a harmonisation challenge because the Census Bureau redrew PUMA boundaries in 2012 based on the 2010 Census, creating a discontinuity between pre-2012 and post-2012 data. To maintain a consistent geographic panel, I use a crosswalk to harmonize 2000-boundary PUMAs with 2010-boundary PUMAs. Each 2000-boundary PUMA is assigned to the 2010-boundary PUMA with which it shares the greatest share of its 2010 population. Pre-2012 PUMAs with a population match weight below 60% are dropped, and I retain around 80% of all PUMAs. Three New Orleans PUMAs affected by post-Hurricane Katrina boundary changes are collapsed into a single unit. The resulting harmonized geographic identifier, *puma_final*, is used throughout.

A.2 Sample Restrictions

Starting from all foreign-born individuals in the pooled ACS data, I filter the sample using these restrictions on migrants:

1. **Origin country:** the sample is restricted to migrants from the 20 most populous origin countries in the data, measured by survey-weighted population counts. This covers the large majority of the immigrant population while avoiding identification driven by very small origin-country migrant counts.
2. **Foreign-born:** individuals born outside the United States ($BPL \geq 100$) with a valid year of immigration ($YRIMMIG > 0$).
3. **Wage earners:** for the wage sample, individuals must report positive wage and salary income, usual hours worked per week, and a valid weeks-worked interval. Imputed hourly wages outside

the range 1–500 are dropped to avoid outliers, or implausible hourly wages for wage-earners.

4. **Recently arrived:** the regression sample is further restricted to immigrants who arrived in the survey year or the year prior (years since migration ≤ 1), so that outcomes are measured close to the time of arrival, when migrants face the most labor market frictions and are most likely to take advantage of their networks.
5. **Employment outcomes** For the employment outcome, the sample is constructed separately starting from a broader base that retains non-wage earners. It includes all immigrants in the labor force ($\text{EMPSTAT} \in 1, 2$, employed or unemployed) satisfying the origin-country, foreign-born, and years of migration restrictions above.

A.3 Wage Construction

The ACS does not report exact weeks worked prior to 2016. Instead, it reports only an intervalled weeks-worked variable, `WKSWORK2`. I impute weeks worked using the midpoint of each bracket: 7 weeks for bracket 1, 20 weeks for bracket 2, 33 weeks for bracket 3, 43.5 weeks for bracket 4, 48.5 weeks for bracket 5, and 51 weeks for bracket 6.

Annual hours are computed as usual weekly hours, `UHRSWORK`, multiplied by imputed weeks worked. Hourly wage is defined as:

$$\text{wage_hourly}_i = \frac{\text{INCWAGE}_i}{\text{UHRSWORK}_i \times \widehat{\text{weeks}}_i}. \quad (4)$$

The log hourly wage is the natural logarithm of this variable.

A.4 Endogenous Variable: Coethnic Network Share

The endogenous regressor measures the concentration of coethnic immigrants in a migrant’s PUMA. For each PUMA j , origin country k , and year t , I compute the weighted count of immigrants from country k using IPUMS person weights, `PERWT`. I then divide this count by the total weighted immigrant population in PUMA j in year t across all 20 origin countries:

$$\text{network_share}_{jkt} = \frac{\sum_{i: \text{origin}_i=k, \text{puma}_i=j, \text{year}_i=t} \text{PERWT}_i}{\sum_{i: \text{puma}_i=j, \text{year}_i=t} \text{PERWT}_i}. \quad (5)$$

The denominator, `immig_pop_sumjt`, is the total immigrant population in the PUMA-year cell. It enters the wage regressions as an additional control to separate the composition effect (who is in the PUMA), from the scale effect (how many migrants are in the PUMA).

A.5 Shift-Share Instrument

Base-year shares. For each origin country k and PUMA j , I compute the share of origin- k migrants living in PUMA j relative to all migrants living in PUMA j in the base year, t_0 at 2005.

$$\text{share}_{jk,2005} = \frac{\sum_{i: \text{origin}_i=k, \text{PUMA}_i=j, \text{year}_i=2005} \text{PERWT}_i}{\sum_{i: \text{PUMA}_i=j, \text{year}_i=2005} \text{PERWT}_i}. \quad (6)$$

These shares are fixed at their 2005 values for all subsequent years. They capture the predetermined geographic settlement pattern of each origin group before the analysis period begins. Using a fixed baseline reference period is standard in the literature.

National inflow shift. The shift component is the national weighted count of newly arrived immigrants from origin country k in year t . Newly arrived immigrants are defined as individuals whose year of immigration equals the survey year, so that `YRIMMIG = YEAR`:

$$\text{shift}_{kt} = \sum_{i: \text{origin}_i=k, \text{YRIMMIG}_i=t} \text{PERWT}_i. \quad (7)$$

Instrument. The shift-share instrument for PUMA j , origin country k , and year t is the product of the base-year share and the national inflow:

$$Z_{jkt} = \text{share}_{jk,2005} \times \text{shift}_{kt}. \quad (8)$$

PUMAs where the base-year share is zero (because origin group k had no recorded presence in PUMA j in 2005) are assigned an instrument value of zero.

B Heterogeneity Analysis: Category definitions

B.1 Regions

Canada

Asia: China, Vietnam, Philippines, Korea, India, Japan, Iran

Europe: Yugoslavia, England, Germany, UK ns, Other USSR/Russia, Poland

Latin America: Mexico, Cuba, West Indies, South America, Central America

B.2 Origin Country Income Groups

High income: Canada, England, Germany, UK ns, Korea, Poland, Japan

Upper middle: Other USSR/Russia, China, Yugoslavia, Iran, Cuba, Mexico, West Indies, South America

Lower middle: India, Vietnam, Philippines, Central America, Africa

B.3 Occupations

Table 16: Occupation Group Classification (Census ACS Major Groups)

Label	OCC Codes	Sub-groups
Management, business, science, and arts	10–3550	Management; business/financial operations; computer/mathematical; architecture/engineering; life, physical, and social science; legal; education; healthcare practitioners
Service	3600–4650 [†]	Healthcare support; protective service; food preparation; building/grounds cleaning; personal care and service
Sales and office	4700–5940	Sales and related; office and administrative support
Natural resources, construction, and maintenance	6005–7630 [†]	Farming, fishing, and forestry; construction and extraction; installation, maintenance, and repair
Production, transportation, and material moving	7700–9750 [†]	Production; transportation and material moving
Military	9800–9830	Military-specific occupations
Unclassified (ref.)	0, 9920, other	Not elsewhere classified

[†] Upper bound shifts slightly in 2018+ ACS coding: Service 3601–4655; Natural resources 6005–7640; Production/transport 7700–9760. Source: IPUMS USA ACS occupation classification (usa.ipums.org/usa/volii/occ.acs.shtml).

B.4 Industry

Table 17: Industry Group Classification (IPUMS ACS, NAICS-based)

Label	IND Codes	Coverage
Agriculture, forestry, fishing, and mining	170–290, 370–490	Crop/animal production, forestry, fishing/hunting; mining and support activities
Construction	770	Construction, including specialty trades
Manufacturing	1070–3990	Food, textiles, wood, chemicals, metals, machinery, electronics, transport equipment
Wholesale trade	4070–4590	Motor vehicles, groceries, machinery, miscellaneous wholesale
Retail trade	4670–5790	Automobile dealers, food/beverage stores, general merchandise, nonstore retailers
Transportation, warehousing, and utilities	570–690, 6070–6390	Utilities; air, rail, truck, water transport; couriers and warehousing
Information	6470–6780	Publishing, broadcasting, telecommunications, and other information services
Finance, insurance, real estate, and rental/leasing	6870–7190	Banking, credit, insurance, real estate, rental and leasing
Professional and administrative services	7270–7790	Legal, accounting, engineering, scientific R&D, advertising, consulting, waste management
Education, health care, and social assistance	7860–8470	Education; hospitals, outpatient care, nursing, social assistance, child day care
Arts, entertainment, accommodation, and food	8560–8690	Arts, sports, museums, amusement parks, hotels, restaurants, drinking establishments
Other services (ref.)	8770–9290	Repair, personal care, laundry, religious organizations, private households
Public administration	9370–9590	Executive/legislative bodies, justice, public order, national security
Military	9670–9870	U.S. armed forces and military reserves

Note: Industry codes follow the NAICS-based ACS classification from IPUMS USA. “Other services” (8770–9290) is the regression reference category. IND = 0 or 9920 is coded as missing. Source: IPUMS USA industry classifications.

C Heterogeneity Analysis: Binned Scatterplots

C.1 Heterogeneous Effects by Region of Origin

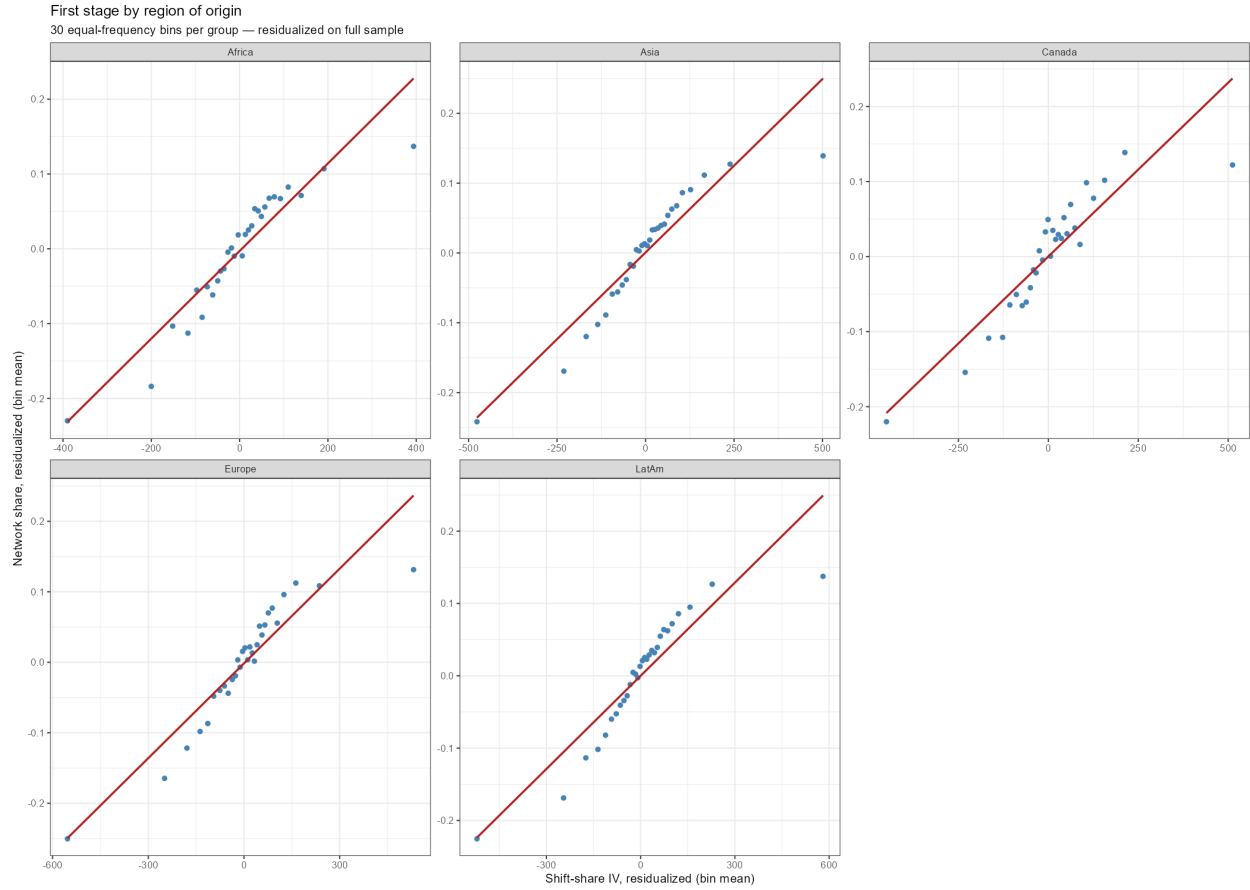


Figure 5: Heterogeneity by Origin

C.2 Heterogeneous Effects by Origin Country's Income Status

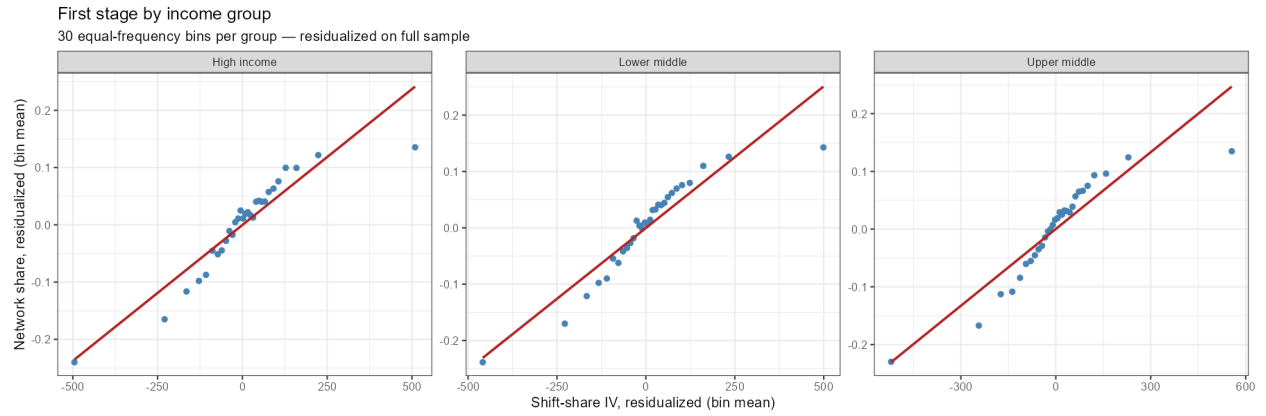


Figure 6: Heterogeneity by Income Status of Country of Origin

C.3 Heterogeneous Effects by Occupation

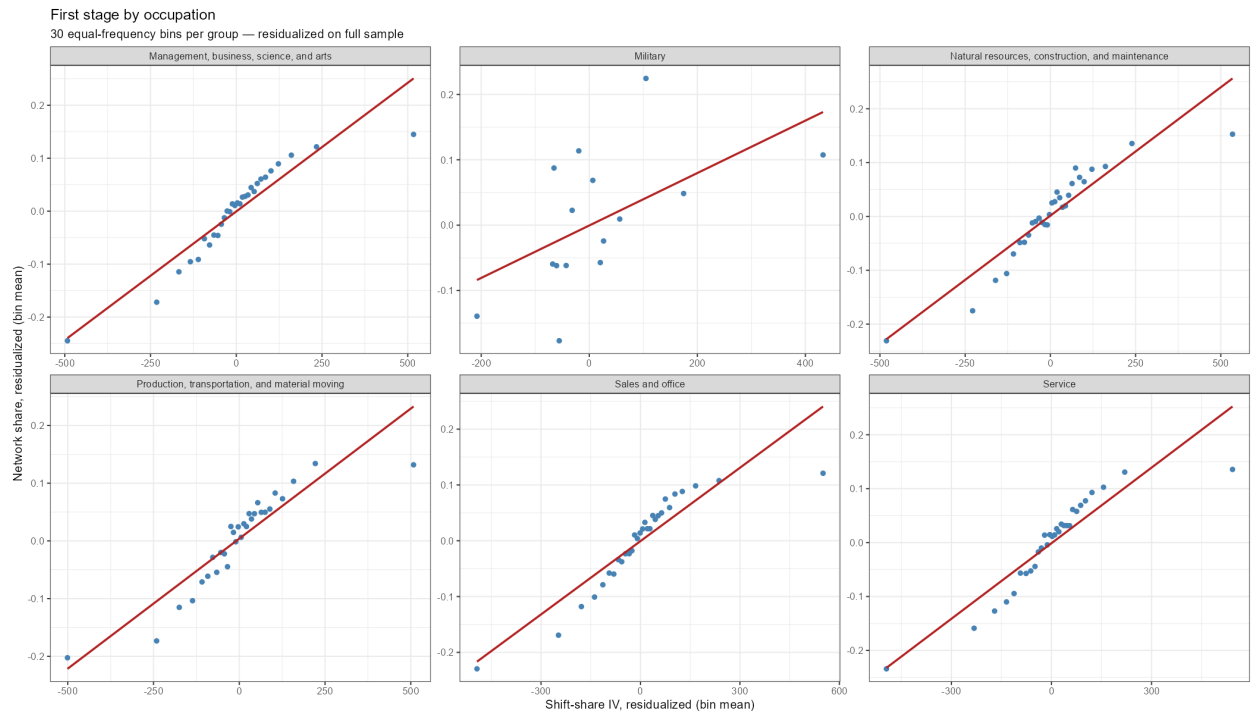


Figure 7: Heterogeneity by Occupation

C.4 Heterogeneous Effects by Industry

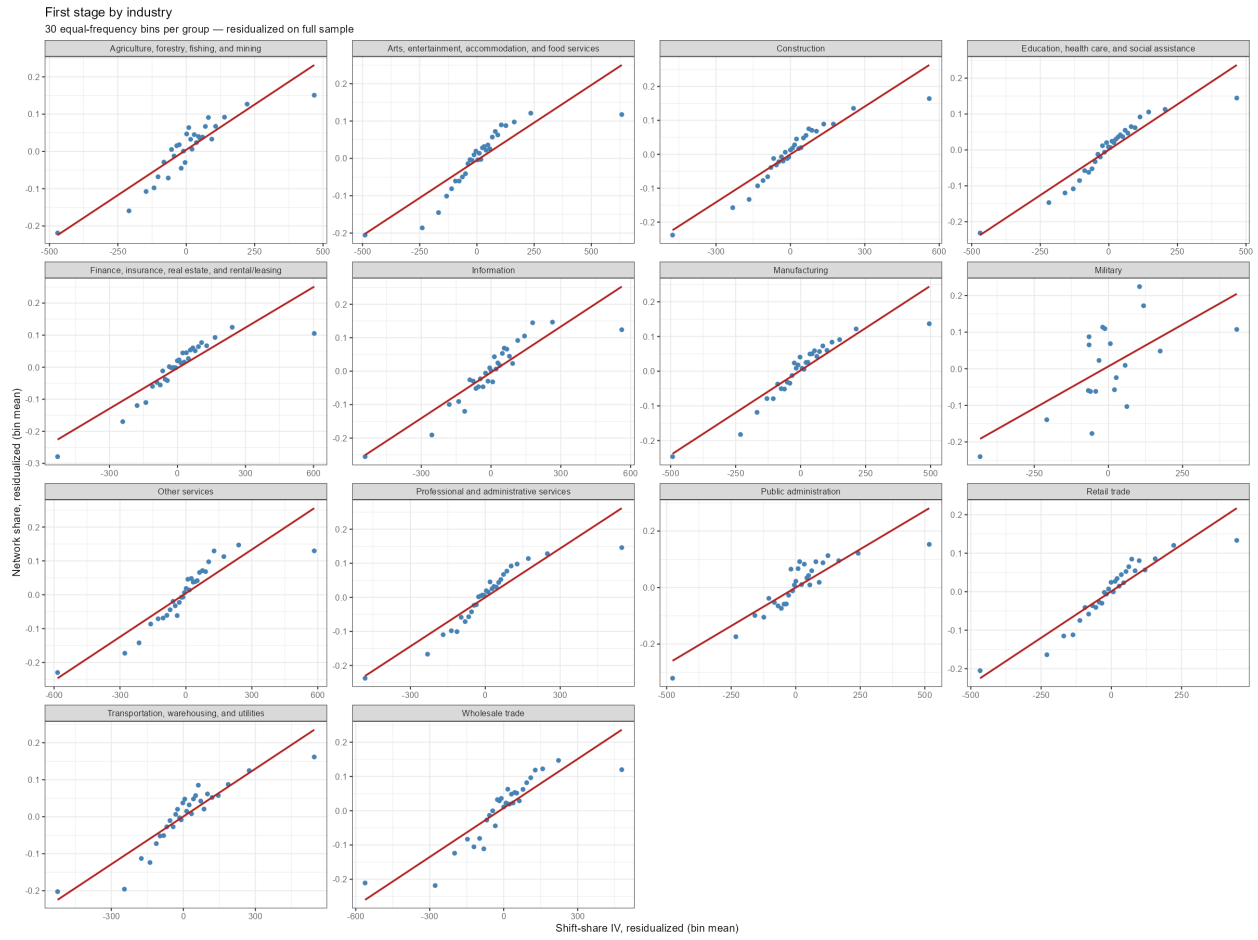


Figure 8: Heterogeneity by Industry

D Discussion of Population Weights

In some regressions, the weighted estimations significantly change the estimates and their distributions. I present here the distribution of the ACS weights used in the weighted estimations.

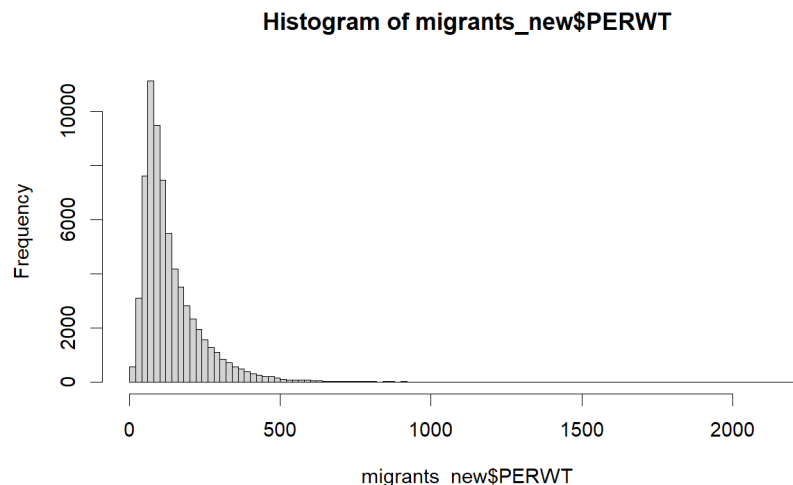


Figure 9: Distribution of ACS person weights

The average PERWT is 137.38, meaning that the average individual in the sample represents approximately 138 people in the population. However, the upper tail of the weight distribution is large. The 75th percentile is 173, but the maximum weight is 2,233. In the weighted regression sample, 29 observations have weights above 1,000. These observations account for only a small share of the 68,388 observations in the sample, but they are weighted equivalent to more than 29,000 observations in the regression. As the results have shown, this would change the estimates and standard errors substantially. It is beyond the scope of this paper to evaluate the representational accuracy of the ACS person weights. However, because large weights can increase the influence of a small number of observations in the weighted regressions, I report both weighted and unweighted specifications in my regression tables. The weights' distribution are as follows:

Mean	137.38
SD	102.58
Min	1.00
P25	71.00
Median	106.00
P75	173.00
Max	2,233.00

Figure 10 and table 18 present the distribution of hourly wage for these 29 migrants whose weights in the ACS are 1000 or higher. The distribution does not seem to contain extreme outliers or any value that would skew the data severely on its own. However, these observations are weighted very disproportionately, more than 1000 times in the regression, and they would affect the estimated coefficient and its distribution.

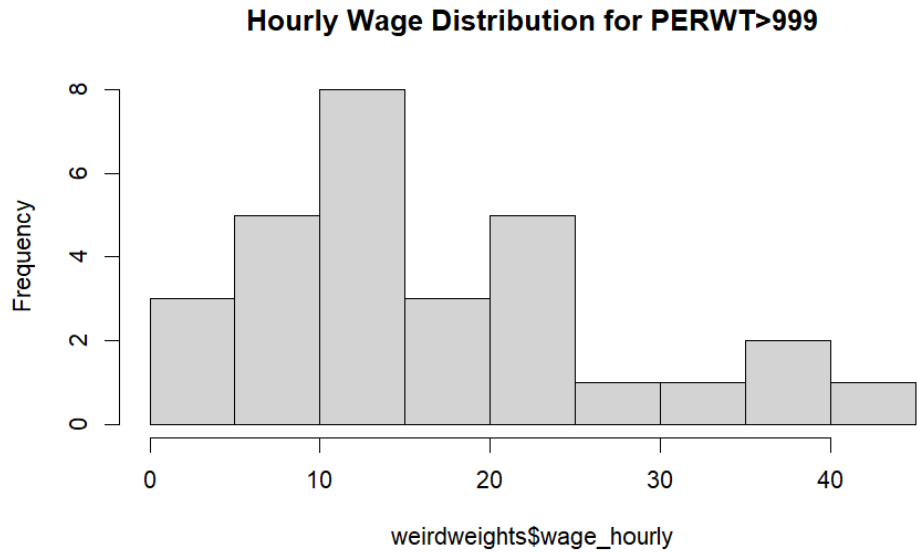


Figure 10: Distribution of Hourly Wage for Highly Weighted Migrants

Table 18: Summary Statistics for Hourly Wage Among Migrants with Weights Above 999

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Hourly wage	3.571	8.824	14.902	16.949	22.059	41.797

E Descriptive Statistics: Tables and Figures

E.1 Hourly Wage Distribution in the Regression Sample

The distribution of hourly wages has a heavy upper tail, as seen in the large difference between the 75th percentile and the maximum. In the high-weight subsample, the 75th percentile is 22.06, while the maximum is 41.80. The difference between this comprehensive, representative sample and the high-weight sample shows a small number of higher-wage observations is in the upper tail distribution.

Table 19: Distribution of Hourly Wage

Statistic	Value
N	68,388.00
Mean	21.71
SD	27.74
Min	1.00
P25	7.45
Median	12.89
P75	26.14
Max	500.00

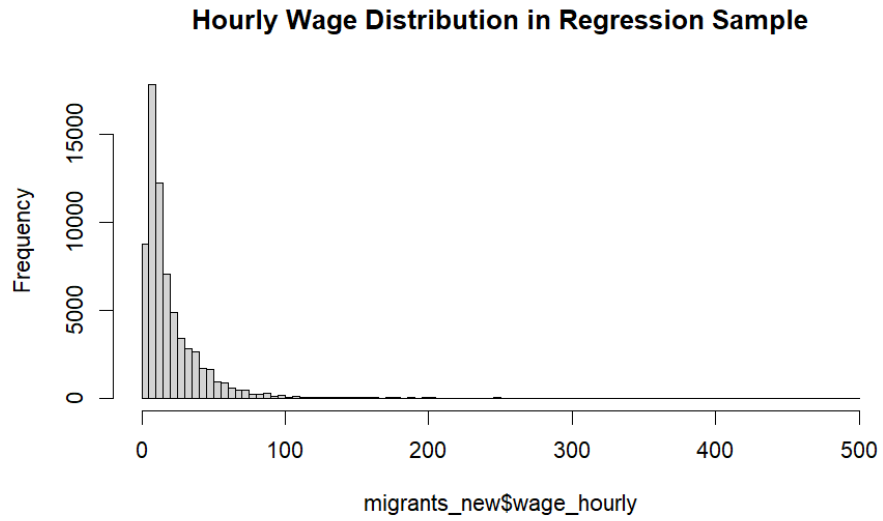


Figure 11: Hourly Wage Distribution

E.2 Distribution of Endogenous Variable - Coethnic Network Share

Figure 12 presents the distribution of the endogenous variable, the coethnic network share, in the regression sample. This distribution shows sufficient variation in across observations, such that the estimation can be sufficiently well powered.

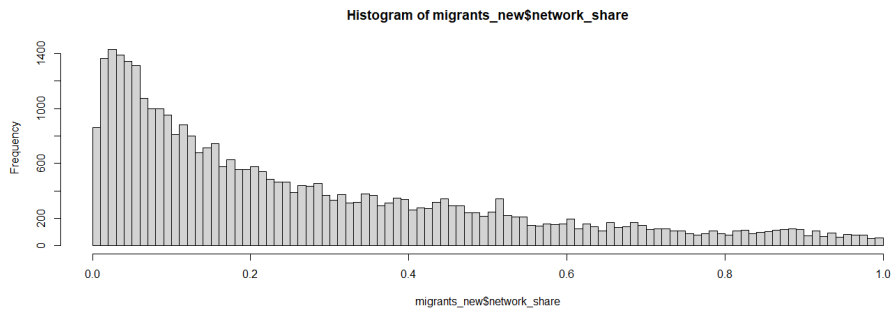


Figure 12: Distribution of Coethnic Network Share

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